

MultiSpec-GeoDiff: Stage-I Demo Report

1. Problem Background

Molecular structure identification from IR spectra is a fundamental inverse problem. This demo proves a runnable Stage-I pipeline:
spectral encoding -> candidate retrieval -> reranking.

2. Dataset

200 experimental NIST IR spectra, split query (1) + library (199).
Interpolated to 400-4000 cm^{-1} grid with min-max normalization.

3. Pipeline

IR spectrum -> interpolate -> normalize -> coordinate-aware encoding (Fourier features + intensity projection) -> distance-bias attention -> global embedding -> cosine similarity retrieval -> forward-spectrum reranking -> Top-K outputs.

4. Core Ideas

- Coordinate-aware encoding: $h_i = W_I I_i + p(\nu_i)$
- Distance bias: $A_{ij} = \text{softmax}(QK^T/\text{sqrt}(d) + b(|\nu_i - \nu_j|))$
- Retrieval: cosine similarity over spectral embeddings
- Reranking: cosine + L1 + peak match scores

5. Outputs: CSV, spectrum overlay, attention heatmap, molecule grid, ablation results, trace log.

6. Limitations

- 200-molecule library, not benchmark-grade
 - No graph diffusion (Stage-II stub)
 - No TFN-Transformer (Stage-III stub)
- Encoder uses random projections, not learned parameters

7. Roadmap

Stage-I: Current demo (implemented)
Stage-II: Graph diffusion + pairwise distance (stub)
Stage-III: TFN-Transformer refinement (stub)